

10 (a)	compares the potentials/voltages at the (inverting and non-inverting) <u>inputs</u>	B1	
	<i>either</i> output (potential) dependent on which input is the larger		
	<i>or</i> $V^+ > V^-$, then V_{OUT} is positive	B1	
	states the other condition	B1	[3]
(b) (i)	ring drawn around both the LEDs (and series resistors)	B1	[1]
(ii)	$V^- = (1.5 \times 2.4)/(1.2 + 2.4) = 1.0\text{ V}$ (allow $1.5 \times 2.4/3.6 = 1.0\text{ V}$)	B1	[1]
(iii) 1.	V_{OUT} switches at $+1.0\text{ V}$ maximum V_{OUT} is 5.0 V when curve is above $+1.0\text{ V}$, V_{OUT} is negative (<i>or v.v.</i>)	B1 B1 B1	[3]
2.	at time t_1 , diode R is emitting light, diode G is not emitting at time t_2 , diode R is not emitting, diode G is emitting (<i>must be consistent with graph line. If no graph line then 0/2</i>)	B1 B1	[2]